Universidad de Costa Rica

Escuela de Ciencias de la Computación e Informática

CI-0146 Compilers

Regular Expressions (RE) - Exercises

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Exercise 12

Description: The set of strings over $\{a, b, c\}$ in which all the a's precede the b's, which in turn precede the c's. It is possible that there are no a's, b's, or c's.

Regular Expression:

$$a^*b^*c^*$$

Explanation: Generates any sequence of a's followed by any sequence of b's, followed by any sequence of c's. It permits the empty string ε and cases where any symbol is absent (e.g., ac, since $b^0 = \varepsilon$).

Exercise 13

Description: The same set as Exercise 12 without the null string.

Regular Expression:

$$aa^*b^*c^* \mid bb^*c^* \mid cc^*$$

Explanation: Equivalent to $a^+b^*c^* \mid b^+c^* \mid c^+$. It generates every non-empty string in $a^*b^*c^*$ by partitioning according to the first occurring character.

Exercise 14

Description: The set of strings of length two or more over $\{a, b\}$ in which all the a's precede the b's.

Regular Expression:

$$aaa^*b^* \mid abb^* \mid bbb^*$$

Explanation: Generates strings in a^*b^* of length $|w| \geq 2$, partitioned into strings with at least two a's, strings with exactly one a and at least one b, or strings with zero a's and at least two b's.

Exercise 15

Description: The set of strings over $\{a, b\}$ that contain the substring aa and the substring bb.

Regular Expression:

$$(a \mid b)^*aa(a \mid b)^*bb(a \mid b)^* \;\mid\; (a \mid b)^*bb(a \mid b)^*aa(a \mid b)^*$$

Explanation: Since aa and bb cannot overlap, they must appear in either order: aa preceding bb, or bb preceding aa.

Exercise 16

Description: The set of strings over $\{a, b\}$ in which the substring aa occurs at least twice. Hint: Beware of the substring aaa.

Regular Expression:

$$(a \mid b)^*aa(a \mid b)^*aa(a \mid b)^* \;\mid\; (a \mid b)^*aaa(a \mid b)^*$$

Explanation: Covers both disjoint occurrences of aa (the first term) and overlapping occurrences within the substring aaa, where aa occurs at indices 1–2 and 2–3.

Exercise 17

Description: The set of strings over $\{a, b, c\}$ that do not contain the substring aa.

Regular Expression:

$$(b \mid c)^*\big(a(b \mid c)(b \mid c)^*\big)^*(a \mid \varepsilon)$$

Explanation: Any occurrence of a must be followed immediately by at least one non-a character (either b or c), except possibly for a single trailing a at the end of the string.

Exercise 18

Description: The set of strings over $\{a, b\}$ that do not begin with the substring aaa.

Regular Expression:

$$\varepsilon \mid a \mid aa \mid (b \mid ab \mid aab)(a \mid b)^*$$

Explanation: A string does not begin with aaa if its prefix of length up to 3 is not aaa. Strings of length < 3 are ε, a, aa, while longer strings must begin with b, ab, or aab, followed by any arbitrary suffix.

Exercise 19

Description: The set of strings over $\{a, b\}$ that do not contain the substring aaa.

Regular Expression:

$$(b \mid ab \mid aab)^*(\varepsilon \mid a \mid aa)$$

Explanation: Blocks of a's between b's can have length at most two. Every non-trailing block of a's is terminated by a b, and the string may end with zero, one, or two a's.

Exercise 20

Description: The set of strings over $\{a, b\}$ that do not contain the substring aba.

Regular Expression:

$$b^*a^*(bbb^*aa^*)^*b^*$$

Explanation: To avoid aba, any transition from an a to a subsequent a via b's must contain at least two consecutive b's (bbb^*). A single isolated b can only appear at the very end of the string or before any a's.

Exercise 21

Description: The set of strings over $\{a, b\}$ in which the substring aa occurs exactly once.

Regular Expression:

$$(b \mid ab)^*aa(b \mid ba)^*$$

Explanation: The unique occurrence of aa is placed in the center. The prefix $(b \mid ab)^*$ ensures that no two a's are adjacent and ends with b (or is empty), preventing aaa. Symmetrically, $(b \mid ba)^*$ ensures no two a's are adjacent to the right and begins with b (or is empty).

Exercise 22

Description: The set of strings over $\{a, b, c\}$ that begin with a, contain exactly two b's, and end with cc.

Regular Expression:

$$a(a \mid c)^*b(a \mid c)^*b(a \mid c)^*cc$$

Explanation: Enforces the initial a, separates the exactly two b's with arbitrary characters from $\{a, c\}$, and terminates with the suffix cc.

Exercise 23

Description: The set of strings over $\{a, b\}$ that contain the substring ab and the substring ba.

Regular Expression:

$$(a \mid b)^*\big(ab(a \mid b)^*ba \;\mid\; ba(a \mid b)^*ab \;\mid\; aba \;\mid\; bab\big)(a \mid b)^*$$

Explanation: Accounts for both non-overlapping occurrences ($ab \ldots ba$ or $ba \ldots ab$) as well as overlapping occurrences in the single substrings aba and bab.

Exercise 24

Description: The set of strings over $\{a, b, c\}$ that contain the substrings aa, bb, and cc.

Regular Expression:

$$\begin{aligned}\Sigma^*\big(&aa\Sigma^*bb\Sigma^*cc \mid aa\Sigma^*cc\Sigma^*bb \mid \\ &bb\Sigma^*aa\Sigma^*cc \mid bb\Sigma^*cc\Sigma^*aa \mid \\ &cc\Sigma^*aa\Sigma^*bb \mid cc\Sigma^*bb\Sigma^*aa\big)\Sigma^*\end{aligned}$$

Explanation: Where $\Sigma = (a \mid b \mid c)$. Because aa, bb, and cc pairwise cannot overlap, they must appear in one of the $3! = 6$ possible permutations.

Exercise 25

Description: The set of strings over $\{a, b, c\}$ in which every b is immediately followed by at least one c.

Regular Expression:

$$(a \mid c \mid bc)^*$$

Explanation: Every occurrence of b is introduced as part of the block bc, ensuring it is immediately followed by a c, while a and c can occur arbitrarily.

Exercise 26

Description: The set of strings over $\{a, b, c\}$ with length three.

Regular Expression:

$$(a \mid b \mid c)(a \mid b \mid c)(a \mid b \mid c)$$

Explanation: Matches any 3-tuple of symbols chosen from the alphabet $\{a, b, c\}$.

Exercise 27

Description: The set of strings over $\{a, b, c\}$ with length less than three.

Regular Expression:

$$\varepsilon \mid (a \mid b \mid c) \mid (a \mid b \mid c)(a \mid b \mid c)$$

Explanation: Explicit union over strings of lengths 0, 1, and 2.

Exercise 28

Description: The set of strings over $\{a, b, c\}$ with length greater than three.

Regular Expression:

$$(a \mid b \mid c)(a \mid b \mid c)(a \mid b \mid c)(a \mid b \mid c)(a \mid b \mid c)^*$$

Explanation: Requires at least four characters, followed by any arbitrary number of symbols.

Exercise 29

Description: The set of strings over $\{a, b\}$ in which the number of a's is divisible by three.

Regular Expression:

$$b^*(ab^*ab^*ab^*)^*$$

Explanation: Every cycle in the repetition contains exactly three a's, with an arbitrary number of b's interleaved between them.

Exercise 30

Description: The set of strings over $\{a, b, c\}$ in which the total number of b's and c's is three.

Regular Expression:

$$a^*(b \mid c)a^*(b \mid c)a^*(b \mid c)a^*$$

Explanation: Matches exactly three occurrences of elements from $\{b, c\}$, interspersed with arbitrarily many a's.

Exercise 31

Description: The set of strings over $\{a, b\}$ in which every a is either immediately preceded or immediately followed by b, for example, $baab$, aba, and b.

Regular Expression:

$$(aba \mid ab \mid ba \mid b)^*$$

Explanation: Any occurrence of a is guarded by an adjacent b on at least one side. The building blocks prevent isolated a's or blocks of three or more consecutive a's (aaa), while allowing valid clusters like $baab$ (via $ba \cdot ab$) and aba.

Exercise 32

Description: The set of strings of odd length over $\{a, b\}$ that contain the substring bb.

Regular Expression:

$$\Sigma^*_{\text{even}} \Sigma bb \Sigma^*_{\text{even}} \;\mid\; \Sigma^*_{\text{even}} bb \Sigma \Sigma^*_{\text{even}}$$

Explanation: Where $\Sigma = (a \mid b)$ and $\Sigma_{\text{even}} = (a \mid b)(a \mid b)$. For a string $w = u \cdot bb \cdot v$ to have odd length, $|u| + |v| + 2$ must be odd, which implies $|u| + |v|$ is odd. Thus, either $|u|$ is odd and $|v|$ is even, or $|u|$ is even and $|v|$ is odd.

Exercise 33

Description: The set of strings of even length over $\{a, b, c\}$ that contain exactly one a.

Regular Expression:

$$\Gamma^*_{\text{even}} \big(a(b \mid c) \mid (b \mid c)a \big) \Gamma^*_{\text{even}}$$

Explanation: Where $\Gamma_{\text{even}} = (b \mid c)(b \mid c)$. A string $w = u \cdot a \cdot v$ with $u, v \in \{b, c\}^*$ has even length if and only if $|u| + |v|$ is odd. Therefore, the single a pairs with an adjacent symbol from $\{b, c\}$ as $a(b \mid c)$ or $(b \mid c)a$, surrounded by even-length strings over $\{b, c\}$.

Exercise 34

Description: The set of strings over $\{a, b, c\}$ with an odd number of occurrences of the substring ab.

Regular Expression:

$$(b \mid a^*c)^* aa^* b (b \mid a^*c)^* \Big((b \mid a^*c)^* aa^* b (b \mid a^*c)^* aa^* b \Big)^* a^*$$

Explanation: An occurrence of ab is produced whenever an a-block is immediately closed by b (aa^*b). Non-matching segments between them are generated by $(b \mid a^*c)^*$. The expression selects an initial ab followed by an even number of additional ab occurrences (pairs), ending in optional trailing a's.

Exercise 35

Description: The set of strings of odd length over $\{a, b\}$ that contain exactly two b's.

Regular Expression:

$$(aa)^*b(aa)^*a(aa)^*b(aa)^* \mid$$
$$a(aa)^*b(aa)^*b(aa)^* \mid$$
$$(aa)^*b(aa)^*b(aa)^*a(aa)^* \mid$$
$$a(aa)^*ba(aa)^*ba(aa)^*$$

Explanation: Strings have the form $a^iba^jba^k$ with length $i + j + k + 2$. The total length is odd if and only if $i + j + k$ is odd. This holds if exactly one of $\{i, j, k\}$ is odd (the first three lines) or all three are odd (the fourth line).

Exercise 36

Description: The set of strings over $\{a, b\}$ with an even number of a's or an odd number of b's.

Regular Expression:

$$b^*(ab^*ab^*)^* \mid a^*ba^*(ba^*ba^*)^*$$

Explanation: Union of two languages: the first term generates strings with an even number of a's ($2k$ occurrences of a interspersed with b's), and the second term generates strings with an odd number of b's ($2k + 1$ occurrences of b interspersed with a's).